

Multivariable Calculus: Lagrange Multipliers & Multiple Integrals

Subject: CALC - Multivariable Calculus & Differential Equations | Author: Prof. Denis Auroux / Gilbert Strang / EduSync
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1. Constrained Optimization via Lagrange Multipliers

To optimize $f(x, y, z)$ subject to constraint $g(x, y, z) = k$, set $\text{grad}(f) = \lambda * \text{grad}(g)$.

The scalar λ represents the Lagrange multiplier (marginal rate of change).

Gradient vector points in direction of greatest instantaneous rate of increase.

2. Double & Triple Integration Transformations

Cartesian double integrals over region R transform into Polar coordinates: $dx dy = r dr d\theta$.

Cylindrical coordinates: $dV = r dr d\theta dz$; Spherical coordinates: $dV = \rho^2 \sin(\phi) d\rho d\phi d\theta$.

Fubini's Theorem permits swapping integration bounds over continuous bounded domains.

3. Vector Calculus & Surface Flux

Green's Theorem converts boundary line integrals into double area integrals.

Stokes Theorem equates the circulation of a vector field along a curve to the surface curl flux.